

AUTOMATED INVOICE ANALYTICS

An End-to-End Data Analytics Project

PREPARED BY

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TOOLS & TECHNOLOGIES

Python | Pandas | NumPy | Matplotlib | MySQL | Microsoft Excel 365 | Microsoft Power BI

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1. Introduction

Invoice data plays a significant role in understanding business performance, customer purchasing behavior, product demand, and overall revenue generation. Every business generates thousands of invoices over time, making manual analysis difficult and time-consuming.

The objective of this project is to transform raw invoice data into meaningful business insights using modern data analytics techniques. The project follows a complete analytics workflow beginning with data cleaning in Python, followed by SQL-based business analysis, dashboard creation in Microsoft Excel, and interactive visualization using Microsoft Power BI.

This end-to-end approach enables businesses to monitor sales performance, identify revenue trends, evaluate customer behavior, and support data-driven decision-making.

2. Business Problem

Many organizations collect invoice data daily; however, raw transactional data alone does not provide meaningful business insights. Some common business challenges include:

- Difficulty in identifying high-revenue cities.
- Lack of visibility into top-selling products.
- Inability to monitor customer purchasing patterns.
- Manual reporting consumes significant time.
- Decision-makers do not receive real-time analytical reports.

To address these challenges, an automated invoice analytics system was developed that performs data cleaning, business analysis, KPI generation, and dashboard visualization.

3. Project Objectives

The primary objectives of this project are:

- To clean and preprocess raw invoice data.
- To identify missing values and improve data quality.
- To generate useful business metrics.
- To analyze sales performance using SQL.
- To build business dashboards using Excel.
- To develop interactive Power BI dashboards.
- To identify revenue-generating locations and products.
- To support business decision-making using visual analytics.

4. Dataset Description

The dataset used in this project was obtained from Kaggle and contains invoice transaction records representing customer purchases.

Dataset Details

Attribute	Description
Dataset Source	Kaggle
Dataset Type	CSV
Total Records	Approximately 10,000
Domain	Retail / Sales Analytics

Important Columns

- Invoice Date
- Customer Name
- Email
- City
- Product ID
- Quantity
- Revenue
- Job
- Final Amount

The dataset was cleaned and enhanced before performing business analysis.

5. Data Cleaning using Python

Python was used to clean, preprocess, and transform the raw dataset before analysis.

Libraries Used

- Pandas
- NumPy
- Matplotlib

Data Cleaning Steps

- Imported CSV dataset.
- Checked missing values.
- Removed unnecessary records.
- Converted invoice date into DateTime format.
- Extracted Year, Month, Day, and Weekday.
- Calculated unit price.
- Calculated GST.
- Applied discount calculations.
- Generated Final Amount column.
- Verified duplicate records.
- Created a clean dataset for further analysis.

The cleaned dataset was then exported as **clean_invoice.csv** for SQL, Excel, and Power BI analysis.

6. SQL Business Analysis

Business analysis was performed using MySQL.

SQL Concepts Used

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- LIMIT
- Aggregate Functions
- String Functions
- Date Functions
- Subqueries

Business Queries Performed

- Top 5 Revenue Generating Cities
- Average Revenue by Job
- Top Selling Products
- Total Quantity Sold
- Revenue Analysis
- Product Performance
- Customer Purchase Analysis

SQL helped convert raw business data into meaningful analytical insights.

7. Excel Dashboard

Microsoft Excel 365 was used to build an interactive dashboard using Pivot Tables, Pivot Charts, and KPI Cards.

Dashboard Components

- Total Revenue
- Total Orders
- Average Order Value
- Highest Bill
- Lowest Bill
- Total Quantity Sold
- Revenue by City
- Revenue by Job
- Top Products
- Monthly Revenue Analysis

The Excel dashboard provides a quick overview of business performance and enables users to analyze key metrics efficiently.



Figure 1: Excel Invoice Sales Dashboard

8. Power BI Dashboard

Microsoft Power BI was used to build an interactive business dashboard.

Dashboard Components

- KPI Cards
- Revenue by City
- Revenue by Job
- Top 10 Products
- Monthly Revenue Trend

Power BI provides dynamic visualizations that help business users monitor performance and make informed decisions.

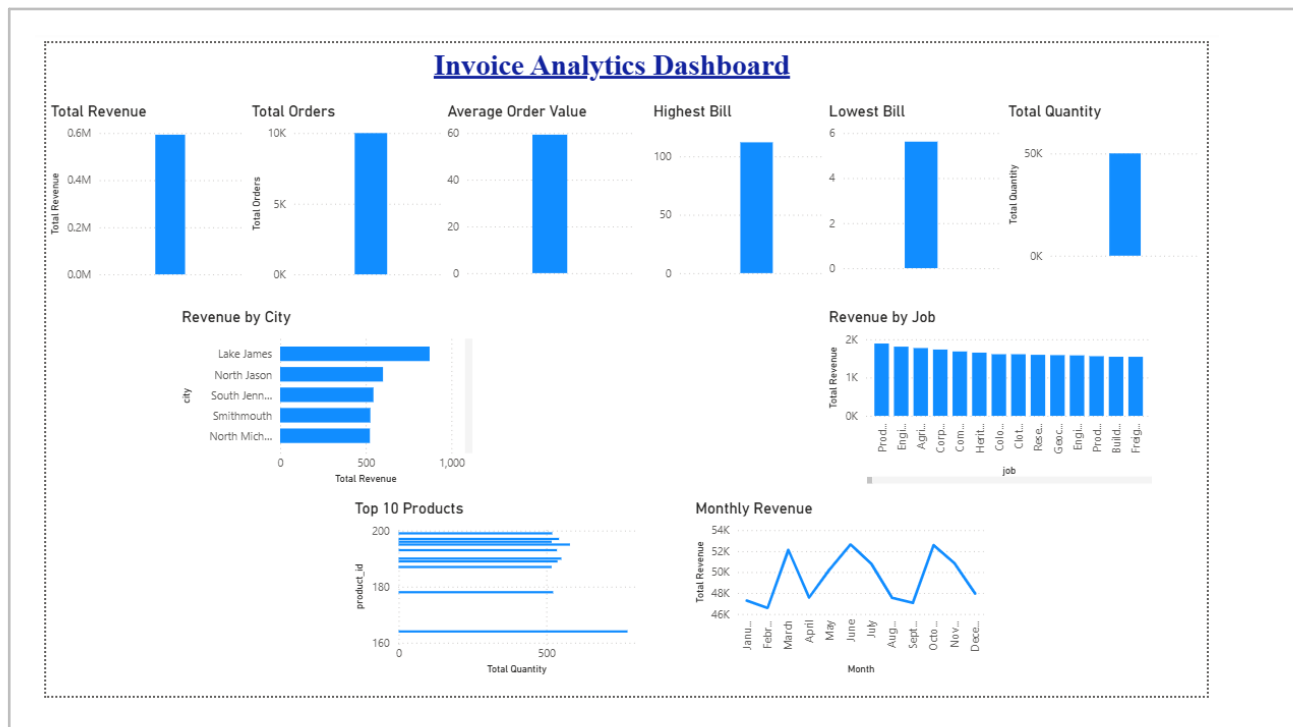


Figure 2: Power BI Invoice Analytics Dashboard

9. Key Performance Indicators (KPIs)

The following KPIs were generated during the project:

KPI	Value
Total Revenue	593,213.23
Total Orders	10,000
Average Order Value	59.32
Highest Bill	112.09
Lowest Bill	5.62
Total Quantity Sold	50,059

These KPIs summarize the overall business performance and provide valuable insights into sales and customer activity.

10. Business Insights

The analysis revealed several important business findings:

- Certain cities generated significantly higher revenue than others.
- A few products contributed a major share of total sales.
- Customer occupation influenced purchasing behavior.
- Monthly revenue trends helped identify periods of high sales.
- KPI analysis enabled quick evaluation of business performance.

These insights can support strategic planning, inventory management, and sales optimization.

11. Technologies Used

Technology	Purpose
Python	Data Cleaning & Preprocessing
MySQL	Business Analysis
Excel 365	Dashboard Development
Power BI	Interactive Visualization
GitHub	Project Version Control

12. Folder Structure

```
Automated_Invoice_Generator/  Dataset/  Documentation/
  Excel/      Images/      PowerBI/      Python/
  SQL/      README.md
```

13. Learning Outcomes

Through this project, the following skills were developed:

- Data Cleaning using Python
- SQL Business Query Writing
- KPI Development
- Dashboard Design in Excel
- Interactive Dashboard Creation using Power BI
- Business Data Interpretation
- End-to-End Data Analytics Workflow

14. Future Scope

The project can be further enhanced by:

- Connecting to live databases.
- Developing real-time dashboards.
- Integrating machine learning models for sales prediction.
- Deploying dashboards on cloud platforms.
- Automating report generation.

15. Conclusion

The Automated Invoice Analytics project successfully transformed raw invoice data into meaningful business insights using Python, SQL, Microsoft Excel, and Microsoft Power BI. The project demonstrates a complete end-to-end data analytics workflow, starting from data preprocessing to business reporting and visualization.

The developed dashboards provide decision-makers with valuable insights into revenue trends, customer behavior, product performance, and sales analysis, enabling more effective and data-driven business decisions.

16. References

- [Kaggle Dataset Repository](#)
- [Python Documentation](#)
- [Pandas Documentation](#)
- [MySQL Documentation](#)
- [Microsoft Excel Documentation](#)
- [Microsoft Power BI Documentation](#)